

Supporting Information for

Fluid inertia controls mineral precipitation and clogging in pore to network-scale flows

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Supporting Information Text

Calculation of Nucleation Rates.

This section outlines the Classical Nucleation Theory (CNT) approach used to estimate the rates of homogeneous and heterogeneous nucleation in a supersaturated barite solution (1, 2). Understanding these rates is critical to predicting the dominant nucleation mechanism. We show that heterogeneous nucleation rate is orders of magnitude higher than homogeneous nucleation rate.

Nucleation rate

The rate of homogeneous nucleation, J , is calculated by the equation:

$$J = \Gamma \exp \left(-\frac{\Delta G_c}{kT} \right)$$

where Γ is the pre-exponential factor, ΔG_c is the free energy change for formation of a nucleus of critical size, k is the Boltzmann constant (1.38×10^{-23} J/K), and T is the absolute temperature (295.15 K).

Critical Free Energy Change

The critical free energy change, ΔG_c , necessary for the formation of a stable nucleus, is defined as:

$$\Delta G_c = \frac{\beta v_0^2 \sigma^3}{(kT \ln \Omega)^2}$$

where β is a geometric factor related to the shape of the nucleus. For a spherical nucleus, β is set to 16.8. v_0 is the molar volume of the solid phase (8.6×10^{-29} m³ for barite). σ is the specific surface energy between the barite crystal and the solution (0.134 J/m²). Ω is the saturation ratio.

Pre-exponential factor

The pre-exponential factor, Γ , is expressed as:

$$\Gamma = 2\pi Z D N_0 N_1 d_c$$

where Z is the Zeldovich factor, D is the diffusion coefficient of BaSO₄ monomers in free water (9.3×10^{-10} m²/s), N_0 is the number of potential nucleation sites per unit volume, set to 3.3×10^{28} for homogeneous nucleation, N_1 is the concentration of BaSO₄ monomers per m³, a value obtained from PHREEQC simulations, and d_c is the critical diameter for nucleation.

Critical diameter and Zeldovich factor

The critical diameter, d_c , is given by:

$$d_c = \frac{4\sigma v_0}{kT \ln \Omega}$$

The Zeldovich factor, Z , is calculated as:

$$Z = \sqrt{\left(\frac{\Delta G_c}{3\pi kT n_c^2} \right)}$$

n_c is the number of monomers within a critical nucleus, given by the equation:

$$n_c = \left(\frac{2\sigma a}{3kT \ln \Omega} \right)^3$$

where a represents the area occupied by a molecule of BaSO₄ (9.42×10^{-19} m²).

Effective surface energy

In the context of heterogeneous nucleation, the surface energy σ is substituted by an effective surface energy σ_{eff} to account for the reduction in the energy barrier due to the presence of a substrate:

$$\sigma_{eff} = \Psi^{\frac{1}{3}}(\theta) \sigma$$

The factor $\Psi(\theta)$, which modifies the surface energy based on the contact angle θ , is expressed as:

$$\Psi(\theta) = \frac{(2 + \cos\theta)(1 - \cos\theta)^2}{4}$$

For heterogeneous nucleation, N_0 is adjusted to represent the concentration of active sites conducive to nucleation, set to 2.5×10^{14} . θ is set to 72.9° (1).

Fig. S2 shows homogeneous and heterogeneous nucleation rates as a function of saturation index.

Vorticity and Z-directional Velocity Analysis.

We examine the normal component of vorticity on the plane a-b-c, as shown in Fig. S6. The vorticity map for $Re = 10$ shows very small vorticity values. This suggests that the vorticity is not significant enough to affect the overall flow pattern. The vorticity map changes as Re increases. The vorticity map for $Re = 150$ has distinct regions of high and low vorticity, suggesting strong rotational flows. The areas of concentrated vorticity coincide with where one would expect Dean vortices to form, and the alternation of red and blue areas suggests that the flow is experiencing these secondary motions. We also investigate the changes of normalized average vorticity magnitude with Re within the entire intersection, as shown in Fig. S8A. The figure shows a clear positive correlation between the Reynolds number and the vorticity magnitude. At the lower end of the Re scale, the vorticity magnitude is low, which is consistent with a creeping flow regime. As Re increases, there is a steady increase in vorticity, rather than a sudden jump, which suggests that the transition from 2D to 3D flow is gradual. This is consistent with the experimental observation that the width of precipitates at the mixing interface gradually increases with Re .

In addition to the vorticity, we also analyze the z-directional velocity and its variability on the plane a-b-c, as shown in Fig. S7. Z-directional velocity is an important quantify of interest because it affects surface accessibility of reactants. At $Re = 10$, the z-directional velocity magnitude is small and spatially uniform, suggesting minimal velocity in the z-direction across the plane, which aligns with a laminar flow profile with little vertical motion. The increase in Re is clearly associated with an increase in the variability and intensity of the z-directional velocity, which indicates the development of three-dimensional flow structures. The most pronounced features are seen at $Re = 150$, where the red and blue regions are well-defined and indicate strong vertical motions, which correlates with the 3D Dean flow. The changes of normalized average z-directional velocity magnitude with Re within the entire intersection are shown in Fig. S8B. It clearly shows the transition from 2D flow ($Re < 10$) to 3D flow as Re increases. The onset of this transition occurs when Re is greater than 10, which matches well with observed precipitation patterns.

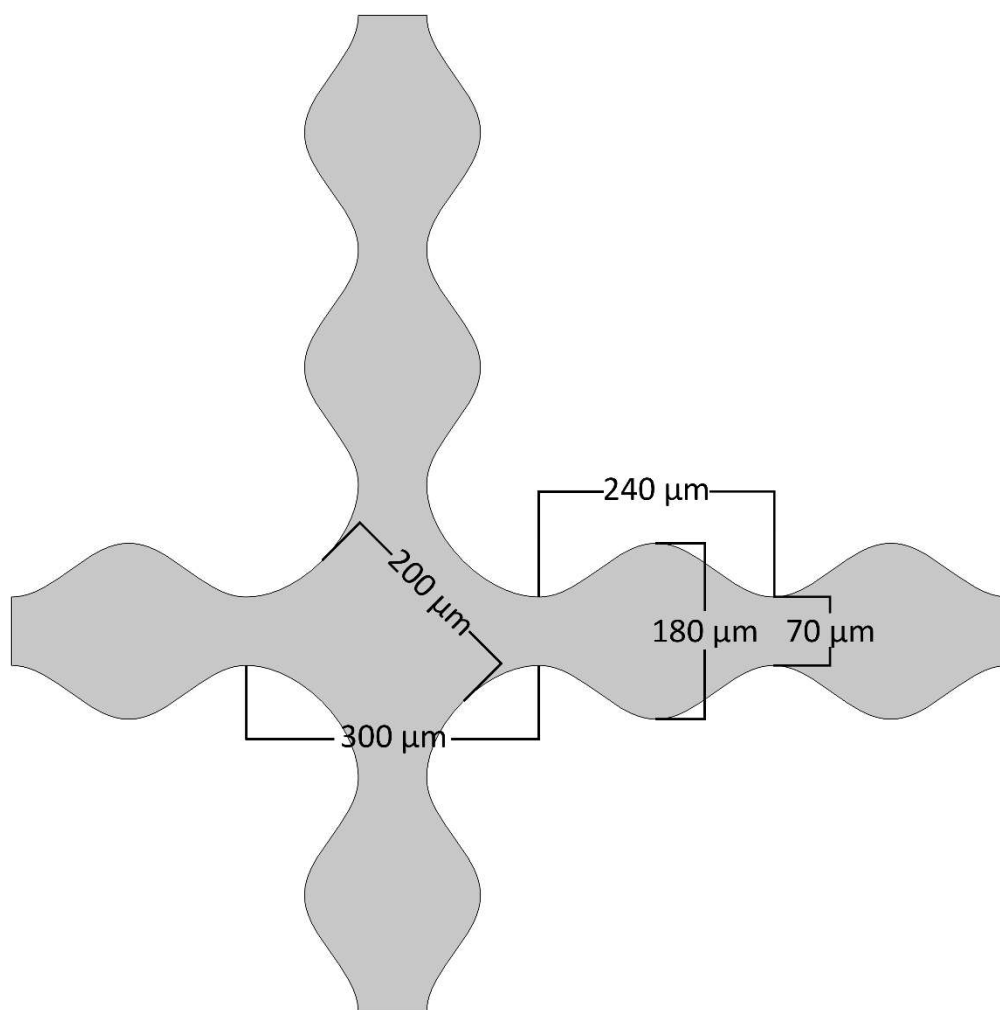


Fig. S1. Schematic of the sinusoidal channel intersection geometry and dimensions.

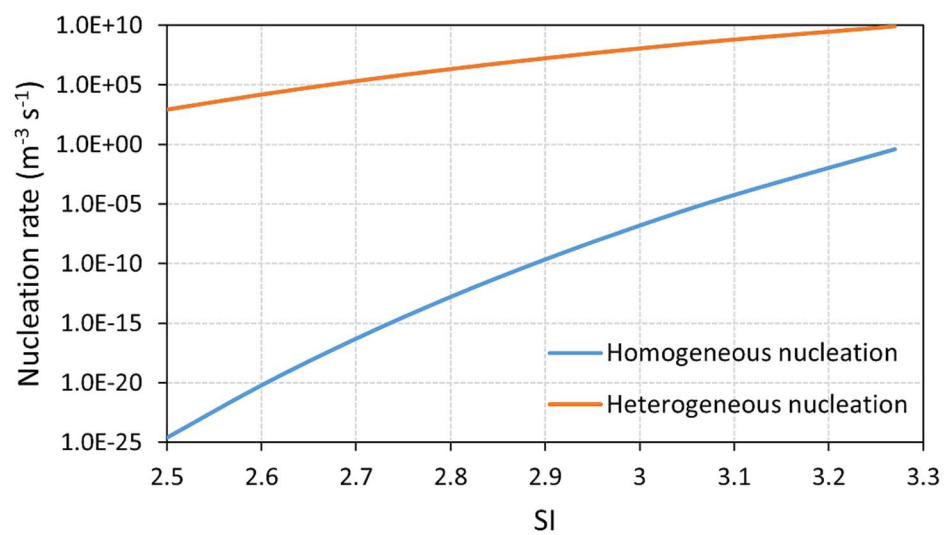


Fig. S2. Homogeneous and heterogeneous nucleation rates as a function of saturation index. Heterogeneous nucleation dominates homogeneous nucleation in the saturation index range of 2.5 to 3.27.

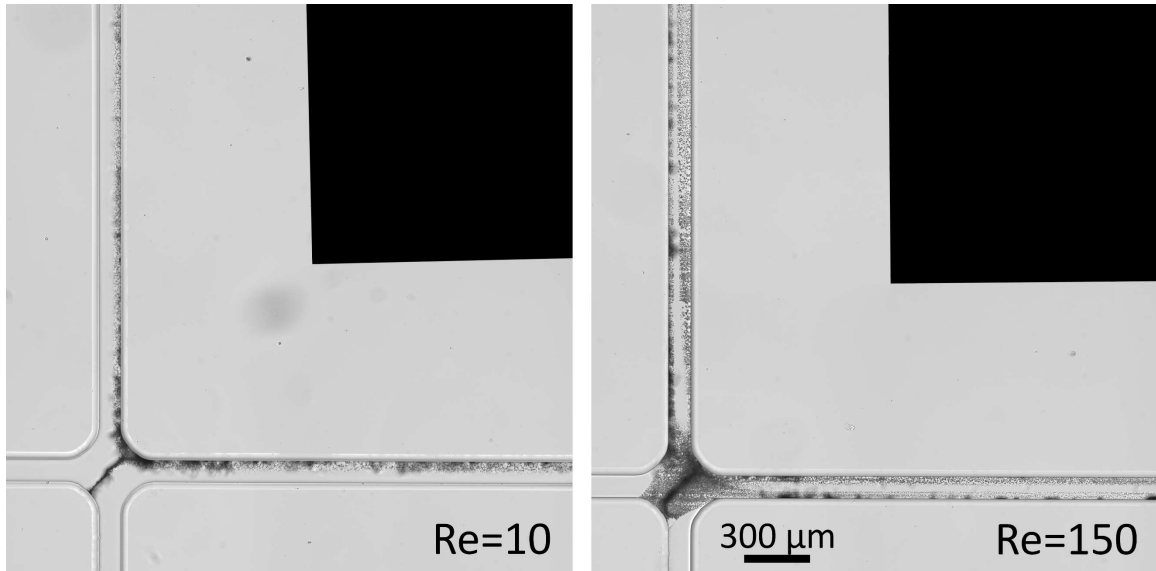


Fig. S3. The patterns of precipitation (shown in black) in straight channel intersections under even flow conditions, obtained from microfluidics experiments at $Re = 10$ and 150 . The width and depth of the channel are 70 and $100\text{ }\mu\text{m}$, respectively. Barium solution was injected from the bottom inlet and the sulfate solution was injected from the left inlet, with $Q_{\text{sulfate}}:Q_{\text{barium}} = 1:1$.

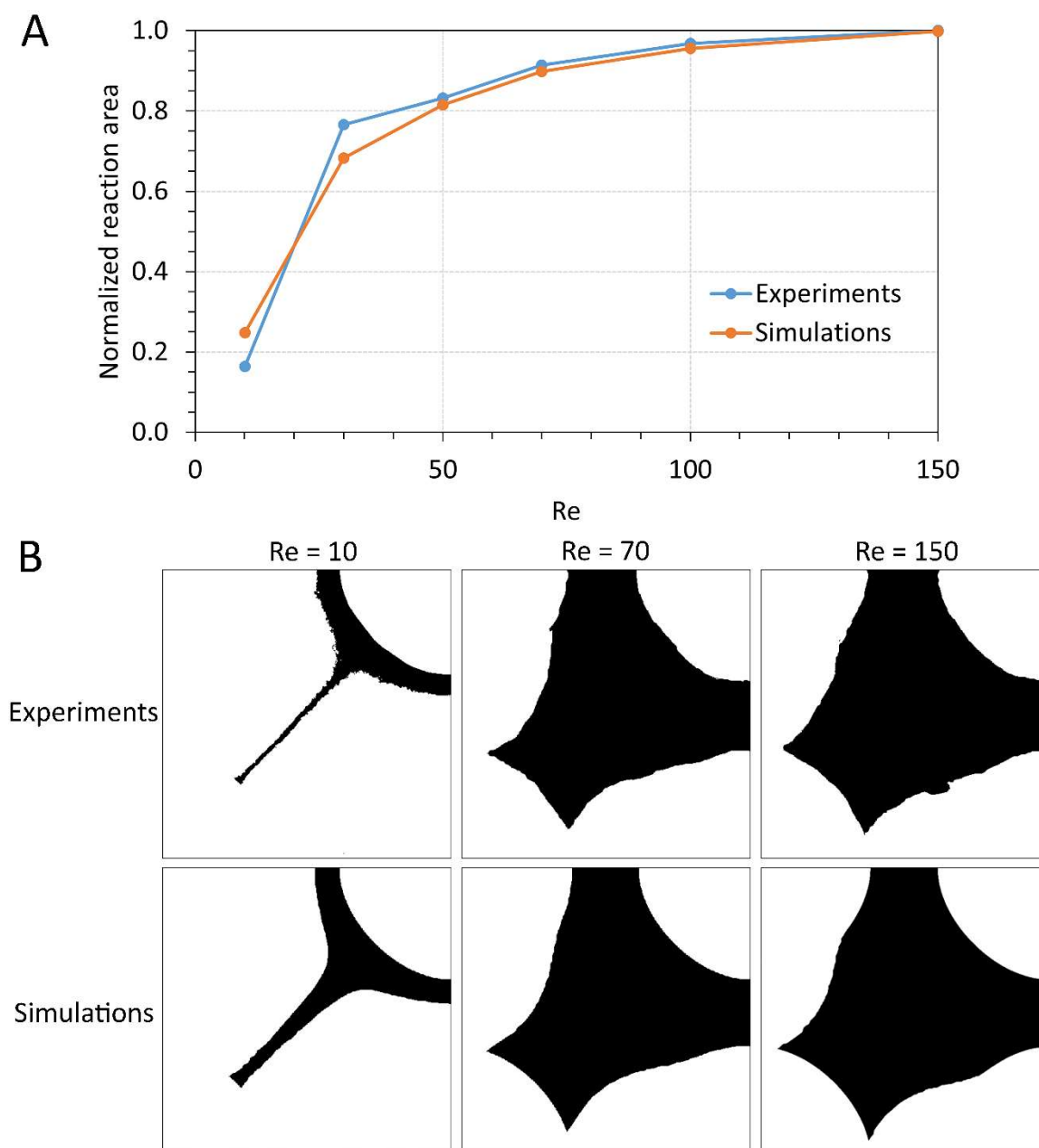


Fig. S4. (A) By processing experimental and simulation images, normalized reaction areas are obtained and plotted as a function of Re . The normalization is based on the experimental surface reaction area at $Re = 150$ with an injection volume of 20 mL. The experimental and simulated reaction areas are calculated from the binarized images, and examples are shown in Fig. S4B. The normalized reaction area matches well between the experiments and simulations confirming the validity and accuracy of the numerical simulations. (B) Binarized images of experimental precipitation patterns and simulated surface reaction areas at different Re values. Otsu's method with auto-thresholding was used for image binarization, and the hole-filling function in ImageJ was applied to the experimental images for a more accurate comparison with the simulation images. The raw experimental and simulation images are shown in Fig. 1. The images clearly show that the surface reaction area enhances significantly as Re increases.

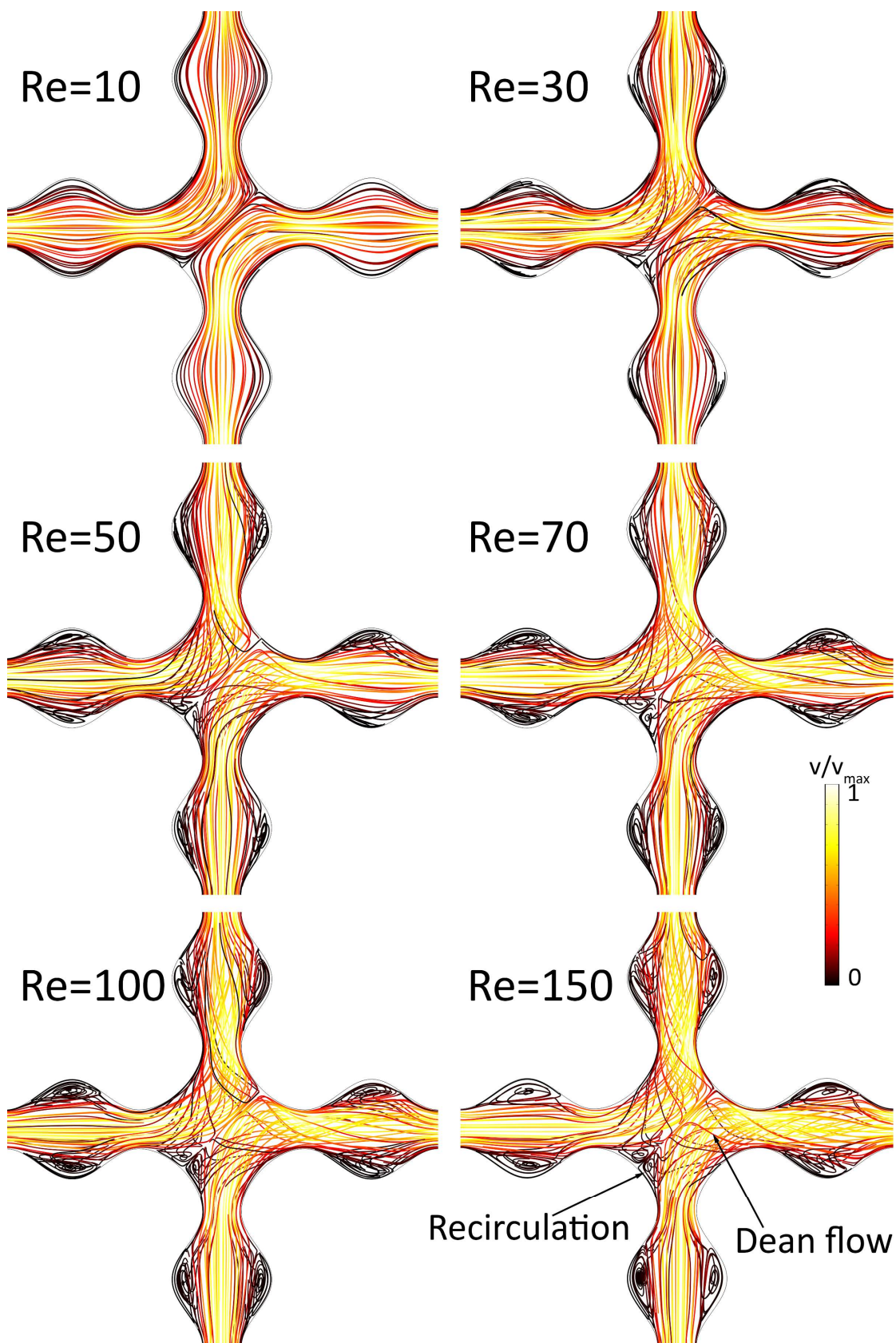


Fig. S5. The streamlines under even flow conditions at different Re values.

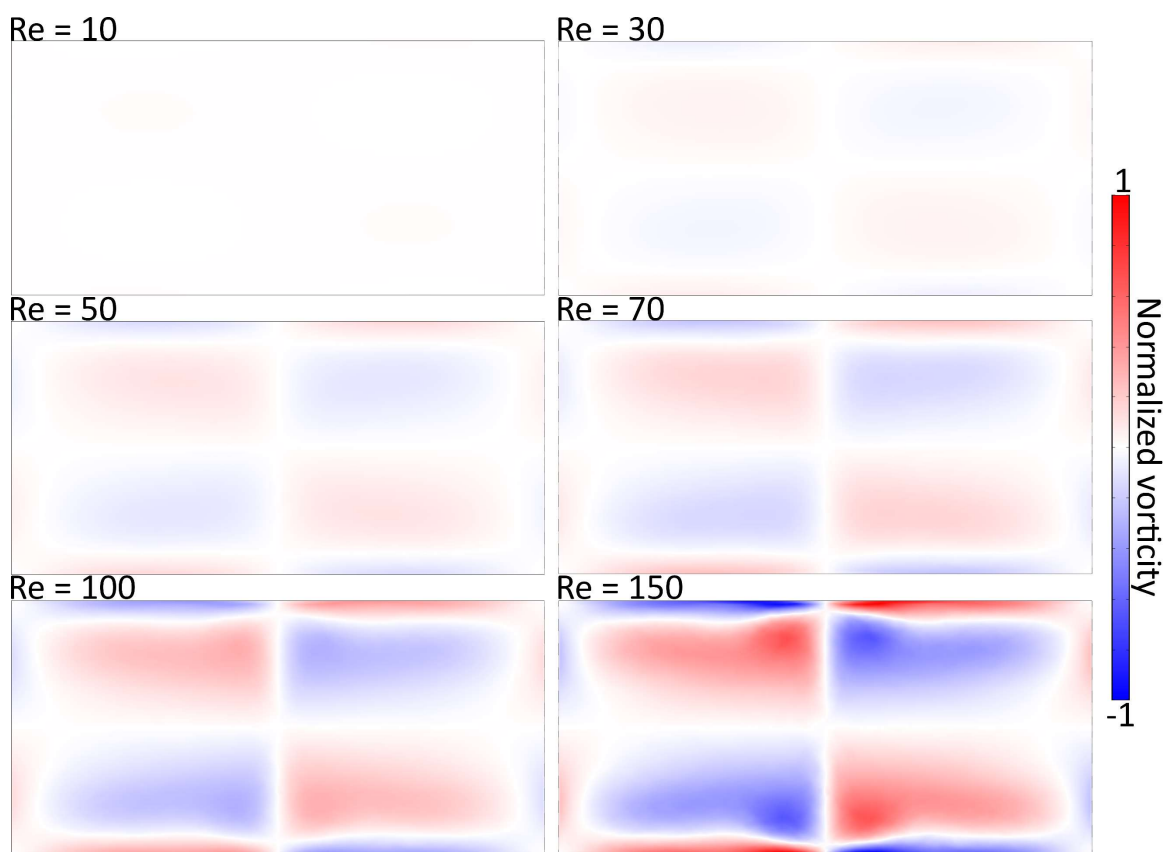


Fig. S6. Maps of normal component of vorticity in the plane a-b-c at different Re values. The color map is normalized to the maximum and minimum values from the Re = 150 case.

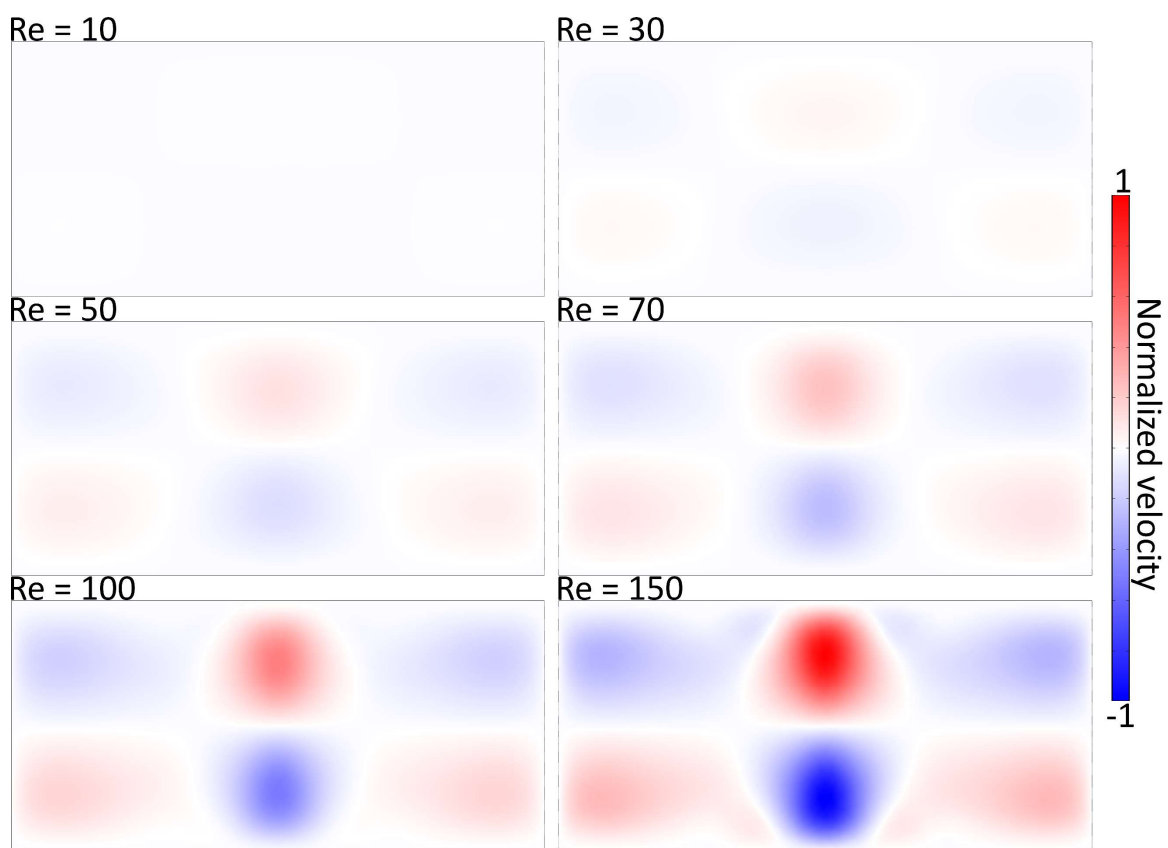


Fig. S7. Maps of z -directional velocity on the plane a - b - c at different Re values. The color map is normalized to the maximum and minimum values from the $Re = 150$ case.

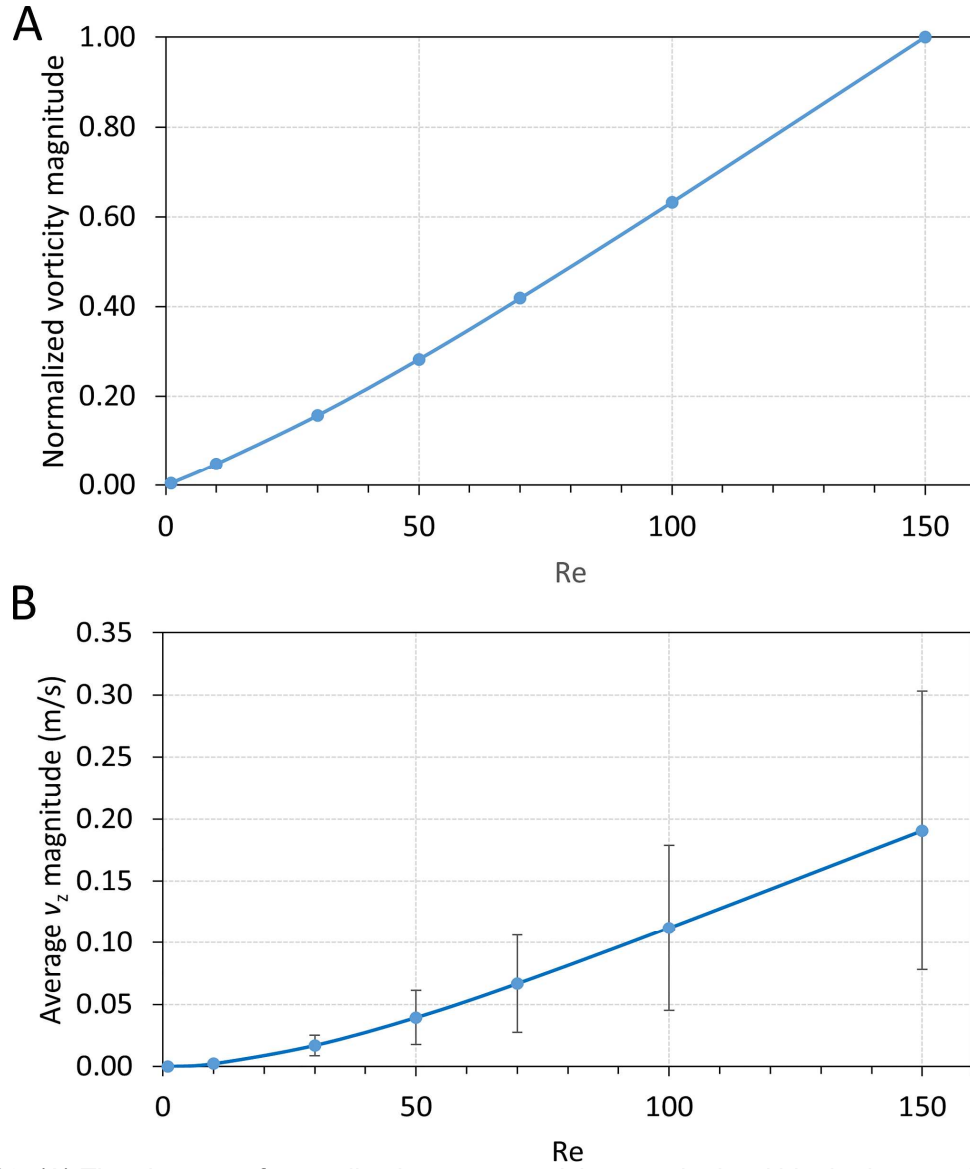


Fig. S8. (A) The changes of normalized average vorticity magnitude within the intersection region (Fig. 2C inset) as a function of Re. (B) The averaged z-directional velocity magnitude within the intersection region as a function of Re. The error bars show half of the standard deviation ($\sigma = \sqrt{\frac{1}{|\Omega|} \int (v_z - \mu(v_z))^2 d\Omega}$), where $|\Omega|$ is the domain volume and $\mu(v)$ is the average z-directional velocity. This quantifies the variability of z-directional velocity. As Re increases, both magnitude and variability of z-directional velocity increase.

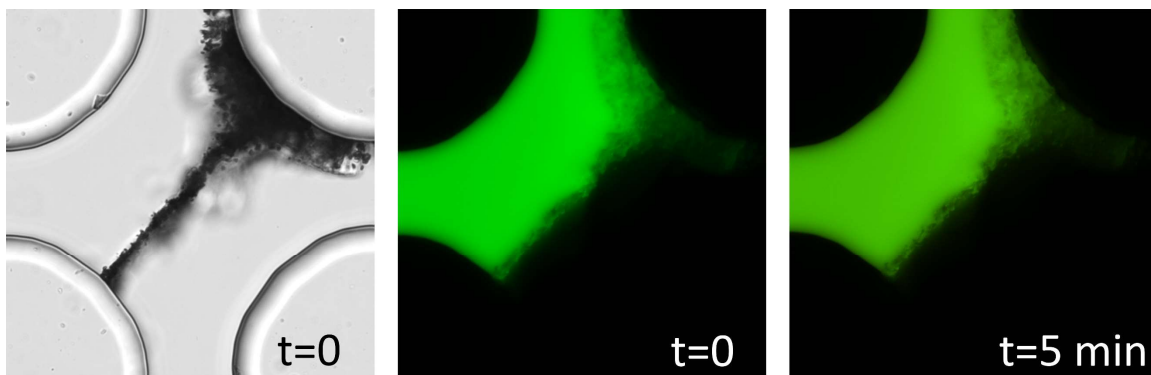


Fig. S9. Following the formation of a dense precipitate layer at $Re = 10$, we stopped co-injection, and solely introduced a sulfate solution containing a 0.1 mM fluorescein tracer at $Re = 0.1$. This was to test if the precipitate layer could hinder further mixing and precipitation. As shown, the fluorescein is initially separated by the precipitate layer at $t = 0$, and even after 5 minutes, there is no discernible transport of fluorescein across the precipitates. This suggests that the precipitate layer is dense and impermeable, effectively terminating any subsequent precipitation. From left to right, the bright field image at $t = 0$, the fluorescence image at $t = 0$, and the fluorescence image at $t = 5$ minutes are shown, respectively.

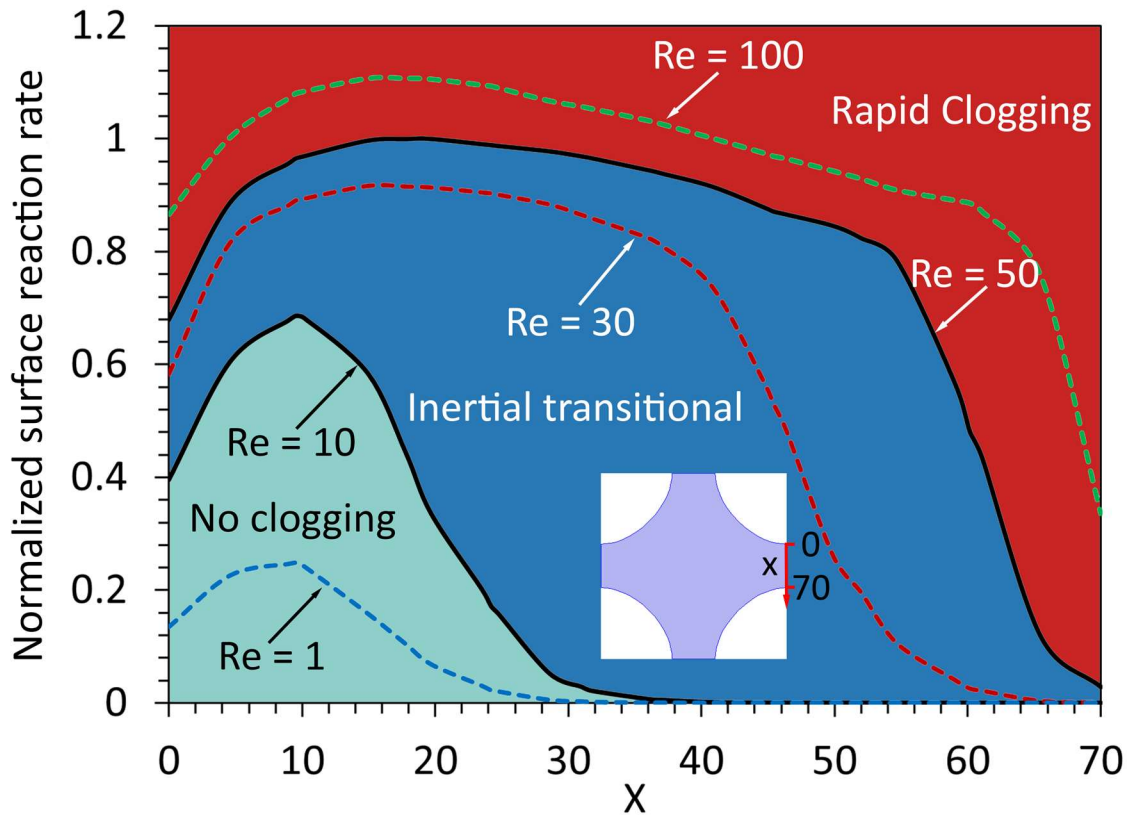


Fig. S10. Criteria used to determine clogging regimes. The normalized surface reaction rate is obtained by normalizing the surface reaction rate with the maximum value of surface reaction rate at $Re = 50$. In the no clogging regime, reactions are confined to one side of the throat, and the reaction zone remains largely unchanged irrespective of Re in the non-inertial laminar regime. The rapid clogging regime is defined as the regime where reaction rate has non-negligible values throughout the entire throat (100%). The inertial transitional regime is marked by an expanded reaction zone, which, however, does not yet cover the entire throat ($< 100\%$). The inset shows the pore throat where the surface reaction rate is evaluated.

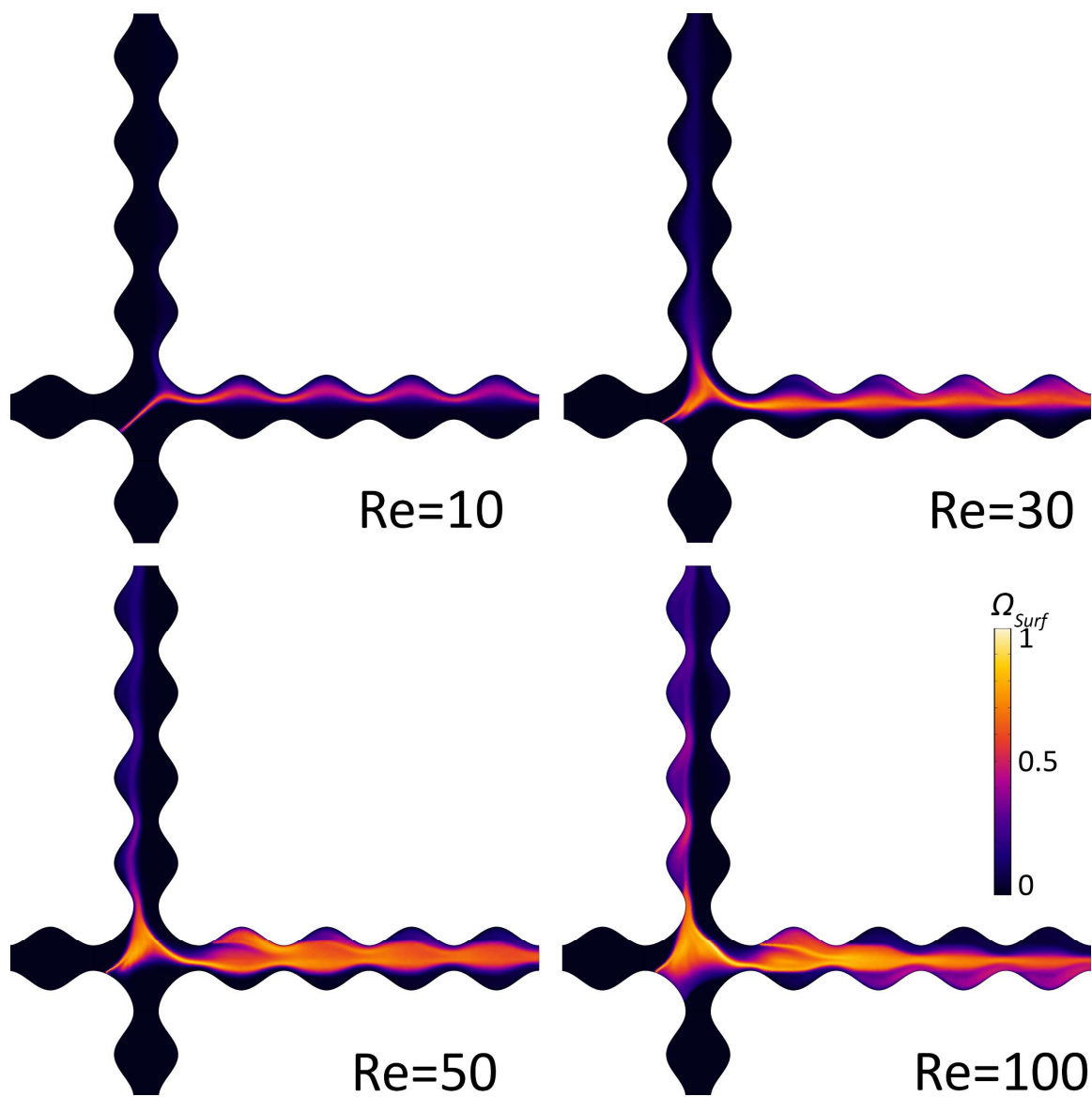
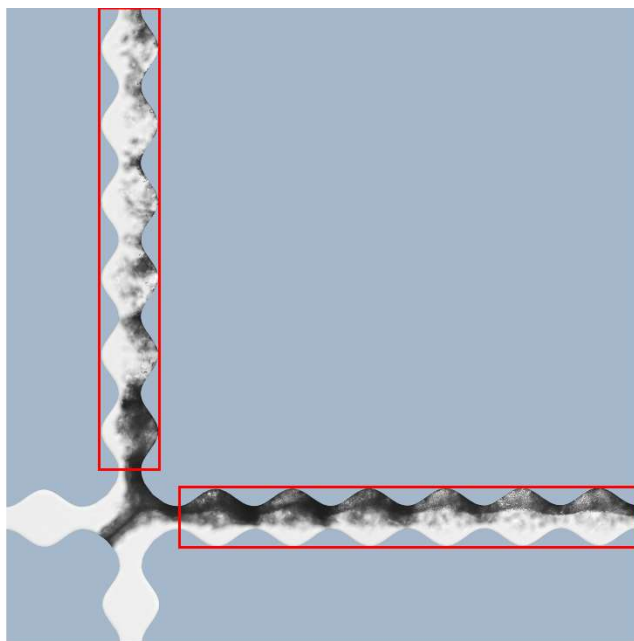


Fig. S11. The spatial Ω_{surf} under uneven flow conditions at different Re values.

A



B

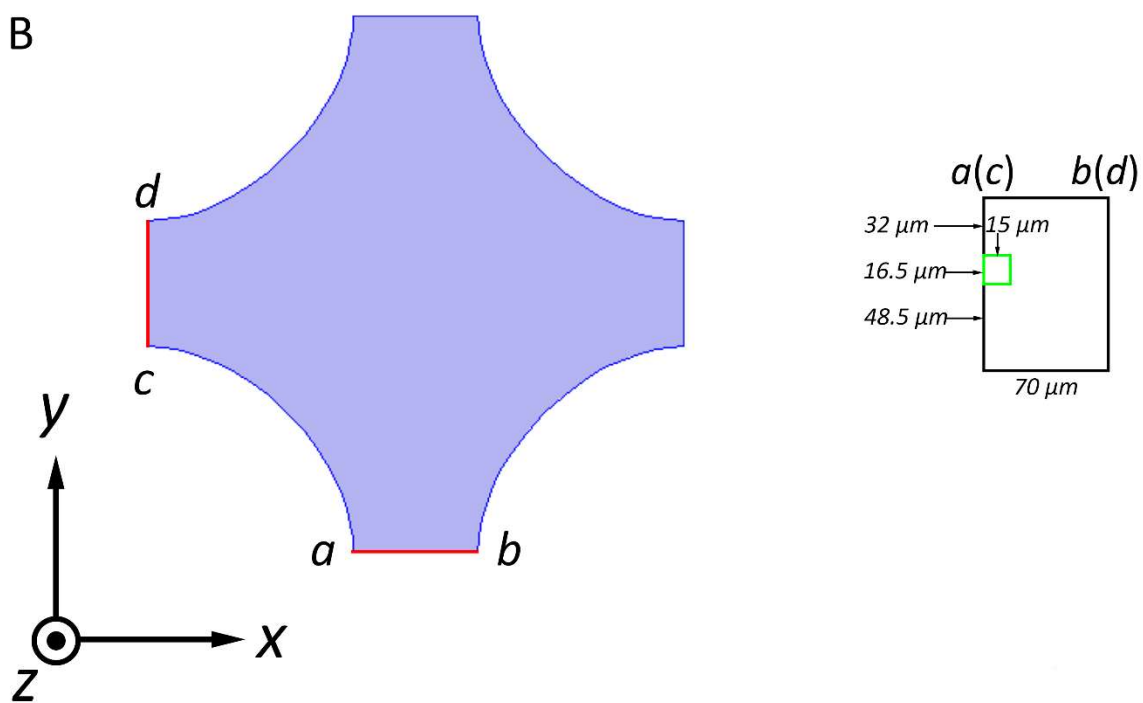


Fig. S12. (A) The regions where the changes in normalized intensity are measured. (B) The green area shows the origin of the streamlines.

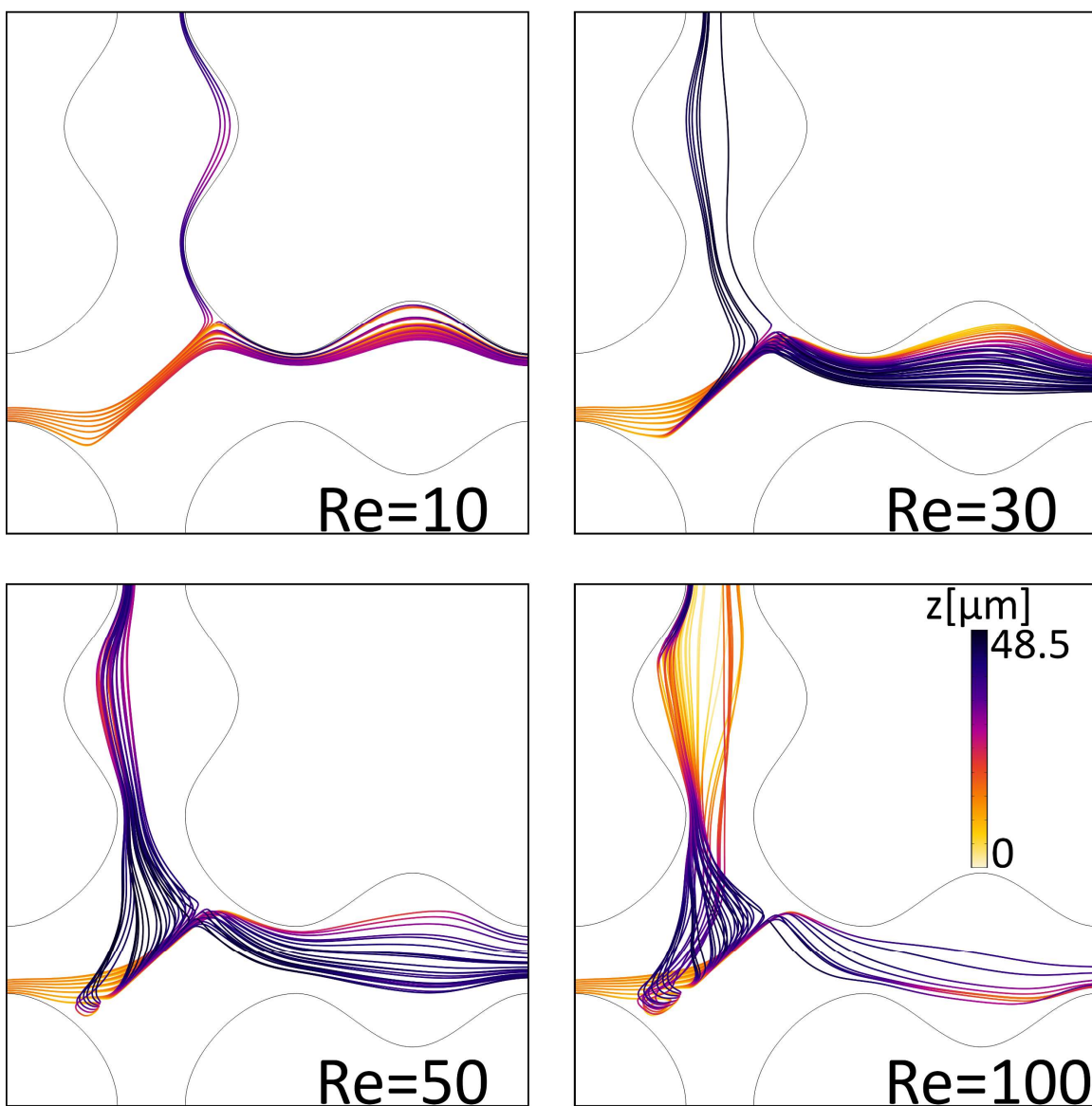


Fig. S13. The reactive streamlines originate from the left inlet under uneven flow conditions for different Re values. Fig. S12B shows the region where the reactive streamlines are initiated.

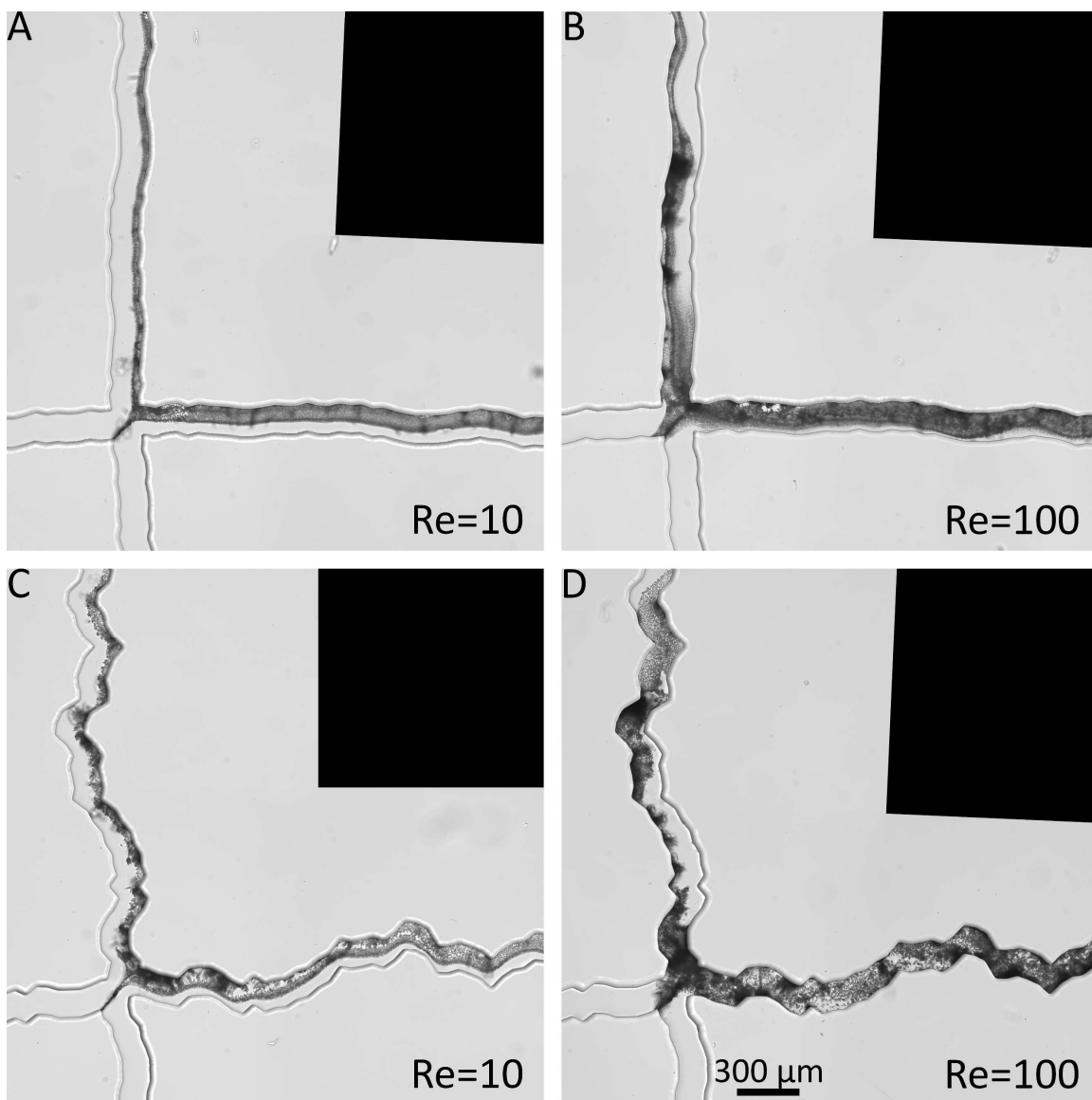


Fig. S14. Precipitation patterns at rough channel intersections for $Re = 10$ and 100 for Hurst exponent value of 0.8 (A and B) and 0.6 (C and D).

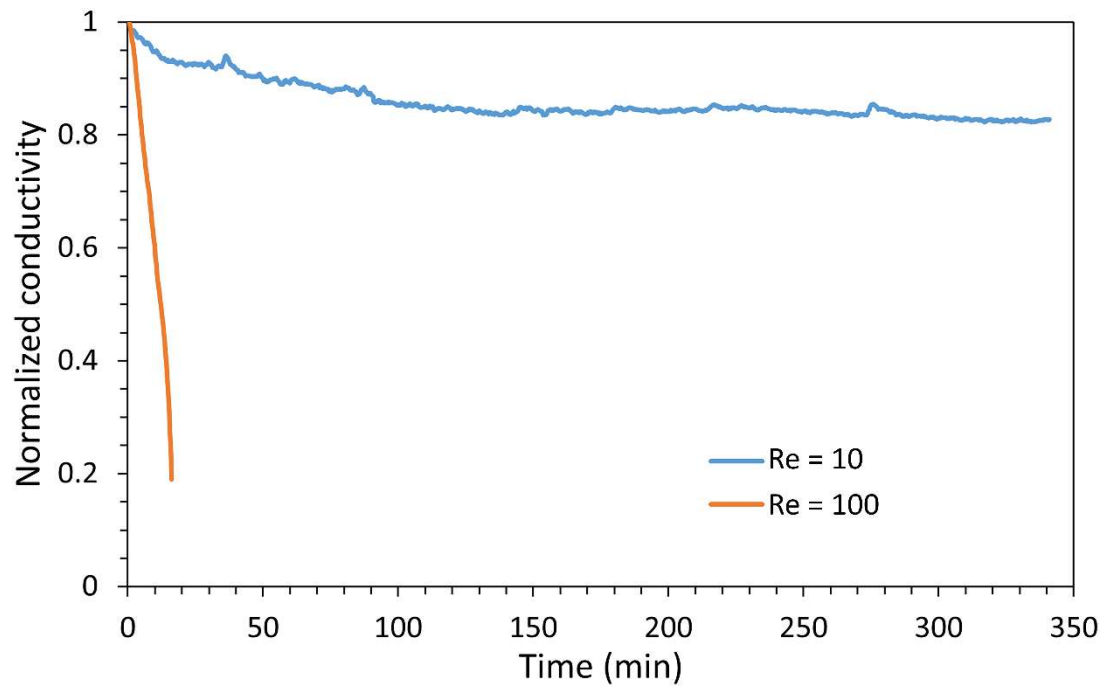


Fig. S15. The evolutions of normalized hydraulic conductivity of channel networks with time for $Re = 10$ and 100 cases.

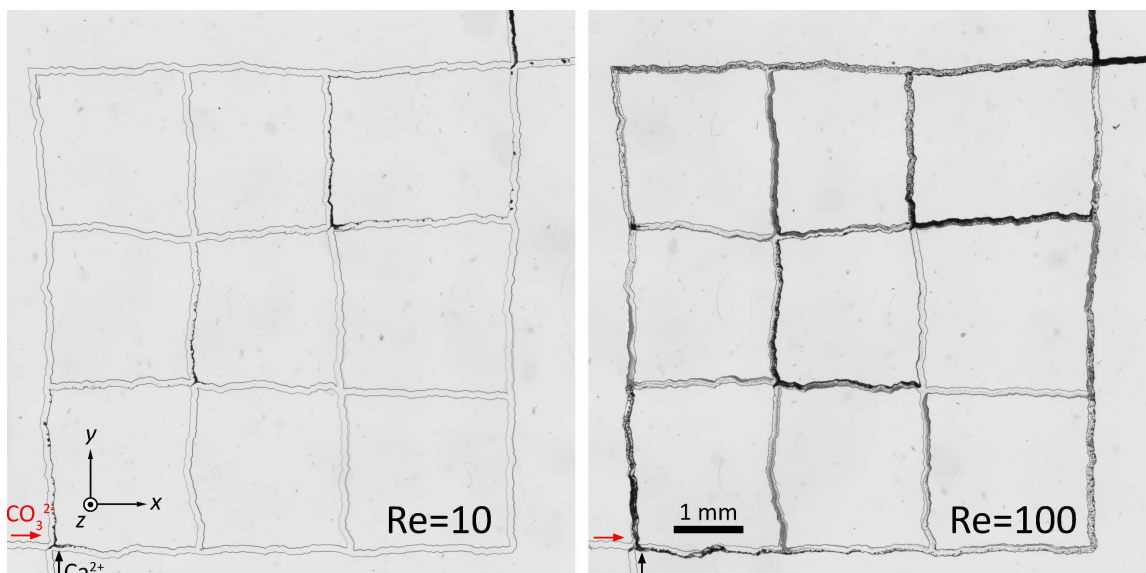


Fig. S16. Calcite precipitation patterns at rough channel networks for $Re = 10$ and 100 . The calcium and carbonate concentrations are 6 mM . The calcium solution was injected from the bottom inlet and the carbonate solution was injected from the left inlet with a flow rate ratio of $1:1$ for the rough intersection and network experiments. The total injection volume is 40 mL .

Movie S1 (separate file). Precipitation dynamics under even flow conditions at $Re = 10, 30$, and 50 .

Movie S2 (separate file). Precipitation dynamics under even flow conditions at $Re = 70, 100$ and 150 .

Movie S3 (separate file). Precipitation dynamics under uneven flow conditions at $Re = 10$ and 30 .

Movie S4 (separate file). Precipitation dynamics under uneven flow conditions at $Re = 50$ and 100 .

Movie S5 (separate file). Barite precipitation dynamics in rough channel networks at $Re = 10$ and 100 .

Movie S6 (separate file). Calcite precipitation dynamics in rough channel networks at $Re = 10$ and 100 .

SI References

1. J. Poonoosamy, *et al.*, Microfluidic investigation of pore-size dependency of barite nucleation. *Commun Chem* **6**, 250 (2023).
2. A. C. Lasaga, "Kinetic Theory in the Earth Sciences" in *Kinetic Theory in the Earth Sciences*, (Princeton University Press, 2014).